

Homework Problems. *Solutions should be written/typed in numerical order and clearly labeled. Unsolved problems should be included but left blank.*

Use the following to answer the next two problems. Let $E = \{1, 2, 3\}$. Consider the following relations in E .

1. State whether each relation below, $\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3, \mathcal{R}_4, \mathcal{R}_5$, is symmetric. Justify your answers.

$$\begin{aligned} \mathcal{R}_1 &= \{(1, 1), (2, 1), (2, 2), (3, 2), (2, 3)\} & \mathcal{R}_4 &= \{(1, 1), (3, 2), (2, 3)\} \\ \mathcal{R}_2 &= \{(1, 1)\} & \mathcal{R}_5 &= E \times E \\ \mathcal{R}_3 &= \{(1, 2)\} \end{aligned}$$

2. State whether each relation below, $\mathcal{R}_6, \mathcal{R}_7, \mathcal{R}_8, \mathcal{R}_9, \mathcal{R}_{10}$, is transitive. Justify your answers.

$$\begin{aligned} \mathcal{R}_6 &= \{(1, 2), (2, 2)\} & \mathcal{R}_9 &= \{(1, 1)\} \\ \mathcal{R}_7 &= \{(1, 2), (2, 3), (1, 3), (2, 1), (1, 1)\} & \mathcal{R}_{10} &= E \times E \\ \mathcal{R}_8 &= \{(1, 2)\} \end{aligned}$$

3. Define the following relation on the real numbers, $\mathcal{R} : x\mathcal{R}y \iff xy \geq 0$. Prove that $\mathcal{R}$ is reflexive and symmetric, but that $\mathcal{R}$ is not an equivalence relation.

4. Define the following equivalence relation on the set of subsets of X as follows: $A\mathcal{R}B \iff A \cap B \neq \emptyset$ for $A, B \subseteq X$. Prove or disprove that $\mathcal{R}$ is an equivalence relation.

5. Let $\mathcal{R}$ and $\mathcal{R}'$ be symmetric relations in a set A . Prove that $\mathcal{R} \cap \mathcal{R}'$ is a symmetric relation in A . (note that $\mathcal{R}$ and $\mathcal{R}'$ are subsets of $A \times A$).

6. Determine the following:

a. $\gcd(2^4 3^2 5 \cdot 7^2, 2 \cdot 3^3 7 \cdot 11)$

b. The lowest common multiple: $\text{lcm}(2^4 3^2 5 \cdot 7^2, 2 \cdot 3^3 7 \cdot 11)$.

7. Determine $51 \bmod 13$, $342 \bmod 85$, and $(82 \cdot 73) \bmod 7$.

8. Find integers s and t so that $1 = 7 \cdot s + 11 \cdot t$. Show that s and t are not unique.

9. Let n be a positive integer greater than 1 and let a and b be positive integers. Prove that $a \bmod n = b \bmod n$ if and only if $a = b \bmod n$.

10. Let $d = \gcd(a, b)$. If $a = da'$ and $b = db'$, show that $\gcd(a', b') = 1$.

Presentation problems.

1. Let the function $f : A \rightarrow B$ be 1-1 and onto, i.e., f^{-1} exists. Prove that $(f^{-1} \circ f) : A \rightarrow A$ is the identity function on A and that $(f \circ f^{-1}) : B \rightarrow B$ is the identity function on B .
 2. Let $f : A \rightarrow B$ be 1-1 and onto. Prove that $g = f^{-1}$, if $(g \circ f) : A \rightarrow A$ is the identity function on A and $(f \circ g) : B \rightarrow B$ is the identity function on B .
 3. The *product set* of sets A and B is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$; it is denoted by $A \times B$. Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
 4. Prove: Let R and R' be symmetric relations in a set A ; then $R \cap R'$ is a symmetric relation in A .
 5. Prove that $\sum_{k=1}^n 2k = n^2 + n$
 6. The Fibonacci numbers are :1, 1, 2, 3, 5, 8, 13, 21, 34, In general, the Fibonacci numbers are defined by $f_1 = 1$, $f_2 = 1$, and $f_{n+2} = f_{n+1} + f_n$ for $n = 1, 2, 3, \dots$. Prove that the n^{th} Fibonacci number $f_n < 2^n$.
 7. Prove that there infinitely many prime numbers.
- ⊙. Two sets are said to be *equal* if there exists a bijective function between them; we write $|A| = |B|$. B is said to have *strictly greater cardinality* than A if there is a one-to-one function from A into B , but there does not exist a bijective function from A into B ; we write $|A| < |B|$. Prove that $|\mathbb{N}| < |\mathbb{R}|$.